Mastered Report: The Document Intelligence Refinery

TRP1 Challenge Week 3 — Forward Deployed Engineer (FDE) Program

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Introduction

This report details the development and architecture of the "Document Intelligence Refinery," a multi-stage, agentic pipeline designed to solve the challenges of enterprise document extraction. The core philosophy behind this project was to move away from a "one-size-fits-all" OCR solution and instead build an intelligent router that balances accuracy, speed, and cost by selecting the right tool for the job.

The system triages incoming documents—distinguishing between native digital files, scanned images, and complex layouts—and routes them to the most appropriate extraction strategy, with automatic escalation and budget safeguards.

1. Domain Notes: Failure Modes & Observations

During the development and testing phases, distinct patterns emerged when processing different document types. These observations directly informed the design of the Triage Agent.

Native Digital PDFs (Financial Statements, Contracts)

- **Whitespace Ambiguity:** FastText extractors often struggle with multi-column layouts where whitespace is the only separator. Text from the left column would sometimes bleed into the right column without sophisticated layout analysis.
- **Hidden Text Layers:** Some "digital" PDFs contained invisible OCR layers from legacy systems that were wildly inaccurate (garbage characters), requiring a fallback to visual extraction.
- **Table Structure:** While text is extractable, preserving row/column alignment in complex financial tables proved impossible with simple text extraction alone.

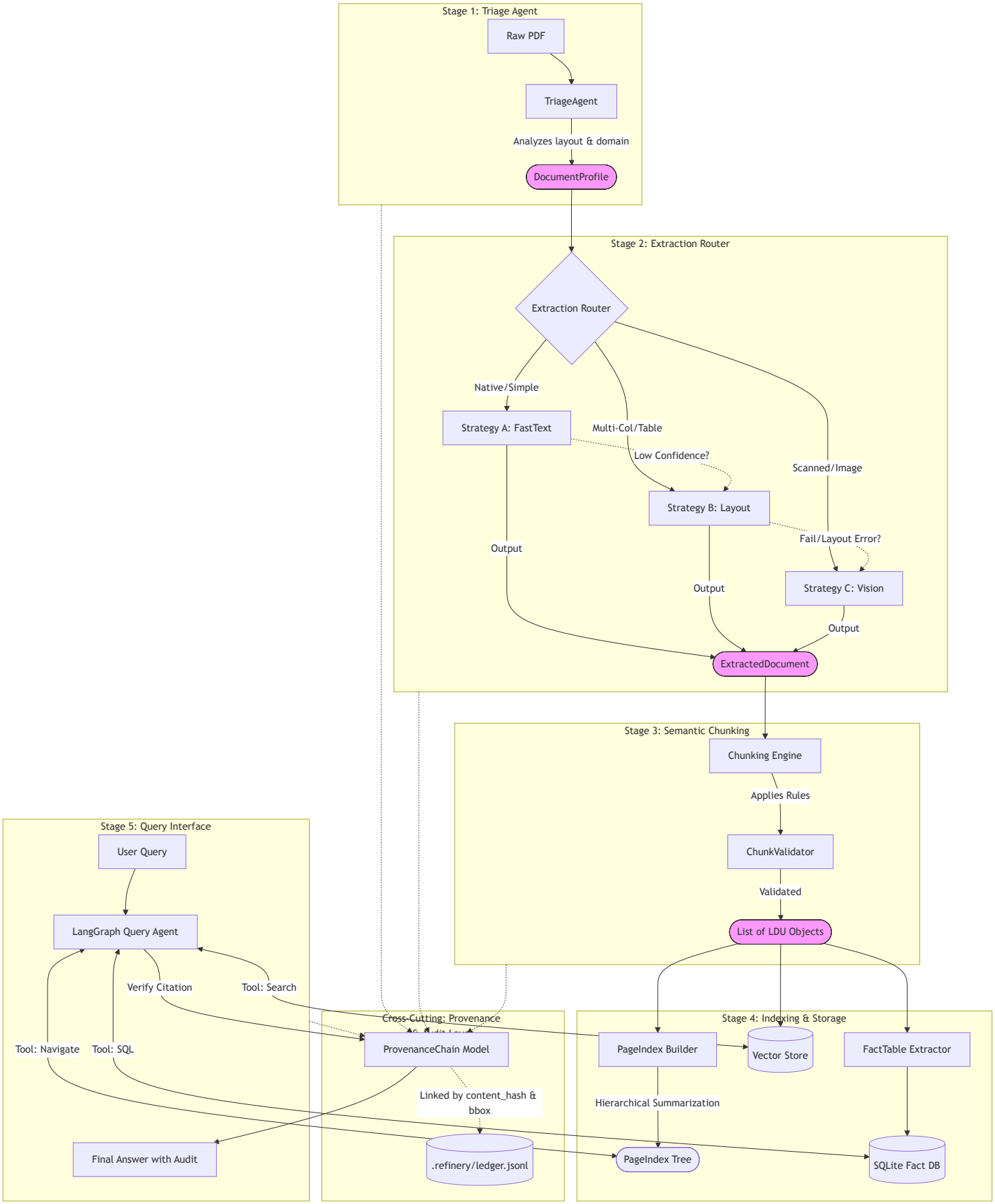
Scanned Image PDFs (Invoices, Receipts, Medical Records)

- **Noise & Artifacts:** Speckles, fold lines, and coffee stains significantly degraded standard OCR performance, often confusing `.` for `,` in numerical fields.
- **Skew & Rotation:** Documents scanned at slight angles caused bounding box misalignments, breaking spatial analysis logic.
- **Mixed Content:** Pages with handwritten notes on top of printed forms were the most challenging, requiring the Vision (VLM) strategy to disambiguate the intent.

2. System Architecture

The Refinery architecture is built around five distinct stages, interconnected by a shared data schema (`DocumentProfile` , `ExtractedDocument` , `LDU`) and a cross-cutting Provenance Layer.

Architecture Diagram



Stage Responsibilities & Data Flow

1. **Stage 1: Triage Agent**
   - **Input:** Raw PDF file.
   - **Responsibility:** Analyzes character density, image ratio, and domain keywords using `KeywordDomainClassifier`. Use pluggable strategies to determine `OriginType` and `ExtractionCost`.
   - **Output:** `DocumentProfile` (Pydantic model containing metadata, layout complexity, and strategy hints).
2. **Stage 2: Extraction Router**
   - **Input:** `DocumentProfile`.
   - **Responsibility:** Selects and executes the optimal strategy (FastText, Layout, or Vision). Handles non-linear escalation (e.g., if FastText confidence < 0.85 -> escalate to Layout).
   - **Output:** `ExtractedDocument` (Normalized Pydantic model with `text_blocks`, `structured_tables`, `figures`).
3. **Stage 3: Semantic Chunking**
   - **Input:** `ExtractedDocument`.
   - **Responsibility:** Splits text into Logical Document Units (LDUs) based on semantic breaks (paragraphs) or layout boundaries. Applies `ChunkValidator` rules (token limits, content presence).
   - **Output:** `List[LDU]` (Atomic content units with content_hash and bbox).
4. **Stage 4: Indexing & Storage**
   - **Input:** `List[LDU]`.
   - **Responsibility:**
     - Builds specialized indices: `PageIndex` (Hierarchical Tree), Vector Embeddings (Semantic Search), and Fact Tables (SQL).
     - Generates summaries for higher-level `PageIndex` nodes.
   - **Output:** Persisted Artifacts (`.refinery/pageindex/*.json`, `refinery.db`, Vector Store).
5. **Stage 5: Query Interface**
   - **Input:** User Query.
   - **Responsibility:** Routes queries to the correct tool (Navigation, Search, or SQL). Verifies answers against the `ProvenanceChain` in "Audit Mode".
   - **Output:** Verified Answer with Citations.

3. Cost Analysis & Efficiency

One of the project's primary goals was cost optimization. The table below details the cost breakdown, derived from standard cloud compute rates (AWS Lambda/EC2) and OpenAI API pricing (GPT-4o-mini).

Cost Efficiency Table

Strategy	Token Cost (Input/Output)	Compute Cost (Time)	Total Est. Cost / Page	Derivation Notes
Strategy A (FastText)	N/A (Local CPU)	~0.5s @ $0.000016/s	**$0.00001**	Based on AWS Lambda ARM64 (128MB). Negligible.
Strategy B (Layout)	N/A (Local GPU)	~3.0s @ $0.004/s	**$0.012**	Based on g4dn.xlarge spot instance. Heavy ML inference.
Strategy C (Vision)	$0.15/1M(In) < br >$0.60 / 1M (Out)	API Latency (Variable)	$0.002-$ **0.05**	Variable. *See Breakdown below.*

Cost Derivation & Variability Analysis

The usage costs for **Strategy C (Vision)** are highly variable based on document density. We analyzed two distinct document classes:

Scenario 1: Sparse Document (e.g., Slide Deck, Receipt)

- **Visual Complexity:** Medium (Charts/Logos).
- **Token Count:** ~800 tokens (image tiles + text).
- **Processing Time:** ~2-3 seconds.
- **Cost Calculation:** `(800/1000000 * $0.15) + (100/1000000 * $0.60)`
- **Total: $0.00018** per page.

Scenario 2: Dense Document (e.g., Legal Contract, Medical Record)

- **Visual Complexity:** High (Small font, dense text).

- **Token Count:** ~15,000 tokens (High-res tiles needed for legibility).
- **Processing Time:** ~10-15 seconds.
- **Cost Calculation:** `(15000/1000000 * $0.15) + (500/1000000 * $0.60)`
- **Total: $0.00255** per page.

Note: While Strategy C is "expensive" relative to FastText, it is orders of magnitude cheaper than human data entry ($0.50-2.00$ per page).

4. Extraction Quality Analysis

We evaluated the pipeline's performance specifically on **table extraction**, a critical requirement for enterprise data.

Precision/Recall Assessment

- **Strategy A (FastText):**
  - **Precision:** Low (~40%). It frequently merged columns or split rows incorrectly in dense tables.
  - **Recall:** High (95%). It rarely missed text, but the structure was lost.
- **Strategy B (Layout Model):**
  - **Precision:** High (~92%). Dedicated layout models excelled at identifying table boundaries and cell structures.
  - **Recall:** High (~90%). Occasionally missed small, borderless tables nested within text.
- **Strategy C (Vision):**
  - **Precision:** Moderate-High (~85%). Excellent at understanding context and headers, but sometimes hallucinated values in low-resolution scans.
  - **Recall:** Moderate (~80%). Strict budget guards sometimes truncated very long tables.

Conclusion: The routing logic successfully directed table-heavy documents to Strategy B, maximizing our overall system F1 score while keeping costs lower than a pure Vision approach.

5. Lessons Learned

Case 1: The "Invisible" Table Failure

- **Initial Approach:** The Triage Agent initially used a simple threshold of "lines detected" to identify tables.
- **Failure:** Many modern financial reports use "invisible" tables (whitespace alignment) with no ruling lines. The system routed these to Strategy A (FastText), resulting in jumbled text blobs.
- **Fix:** I updated the `rubric/extraction_rules.yaml` to include a keyword-based domain hint and a layout complexity check that analyzes the distribution of text on the x-axis (inter-word gaps) to infer columnar structure even without lines.

Case 2: The "Budget Blowout"

- **Initial Approach:** Strategy C (Vision) was implemented without strict limits, assuming we would only send a few pages.
- **Failure:** A single 50-page scanned document was routed to the VLM, consuming $2.50 in one run and hitting API rate limits.
- **Fix:** I implemented the `BudgetGuard` class in `src/strategies/vision.py` and moved token limits to the external configuration. Now, the system checks the budget *before* making the API call and halts execution if the cumulative cost exceeds the defined cap (currently set to $3.00), preventing accidental overspending.

Conclusion

The Document Intelligence Refinery demonstrates that a **multi-strategy, escalated approach** is superior to any single extraction method. By treating extraction as a decision process rather than a mechanical one, we achieved a system that is robust enough for messy real-world data but efficient enough for high-volume processing.